

Muhammad Hammad Irfan

Full-Stack Developer | AI Engineer

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Portfolio | LinkedIn | GitHub

ABOUT

Full-stack developer and AI engineer focused on requirements analysis, system design, and taking SaaS and AI applications from concept to production, translating real-world needs into reliable software.

EDUCATION

National University of Sciences & Technology (NUST)	Islamabad, Pakistan
Bachelor of Science in Computer Science (SEECS)	Sep 2023 – Present

EXPERIENCE

Forward-Deployed AI Engineering Intern <i>Agentic AI Security and Evaluation</i>	DevNeuron - Sunnyvale, CA (Remote) March 2026 – Present
- Lead development of Aginiti-Redteam, an open-source red-teaming framework for enterprise RAG/agent AI systems, implementing data-leakage attack techniques from peer-reviewed research (ICLR 2026, ACM CCS 2025, IEEE TIFS 2026) mapped to the OWASP LLM Top 10. - Engineered a hardened benchmark target simulating real enterprise defenses (RBAC, output redaction, rate limiting) to measure attack efficacy under realistic conditions rather than undefended lab setups. - Designed a provider-agnostic architecture (LiteLLM, ChromaDB) enabling reproducible, black-box red-teaming benchmarks across multiple LLM providers.	
AI and Software Developer (Industry Project) <i>KidneyCare Clinical Assistant</i>	SEECS, NUST June 2025 – Feb 2026
- Engineered a clinical decision support system in collaboration with Kyung Hee University Hospital using rule-based AI for nephrology diagnostics. - Implemented robust data-flow modeling to ensure explainable outputs from unreliable lab datasets.	

PROJECTS

StackShadow – Autonomous AI CTO Co-Pilot
Google Build with AI Hackathon 2026 (GDG on Campus, FAST NUCES)

- Automated infrastructure and cost analysis by building a Next.js/Supabase multi-agent platform.
- Built AI agents via Gemini and GitHub APIs to map dependencies, detect vulnerabilities, and optimize tooling spend.

PKhata – Multi-Tenant Business Management SaaS

- Independently scoped, built, and deployed a production multi-tenant SaaS (Next.js, Supabase) for a real printing business, automating invoicing, customer/order management, payments, and ledger/stock-register tracking with offline-capable workflows.
- Implemented secure organization-level data isolation using Google OAuth and Supabase Row-Level Security to support multiple independent businesses on one platform.

HireMatch – AI-Powered Hiring Match Platform

- Built an applicant tracking system mapping resumes to job descriptions using PGVector for semantic search.
- Developed an LLM pipeline to generate candidate relevance scores and personalized skill gap analyses.

EduSync – A Role-Based University Automation Platform

- Built workflows for scheduling, onboarding, invoicing, and attendance automation.
- Designed RBAC structure using Next.js, Supabase, and Prisma ORM.

LingoMate – An AI-Powered Web Application

- Built an AI conversational app for real-time English fluency practice using React, Azure Speech-to-Text/Text-to-Speech, and the Gemini API.

CORE COMPETENCIES

End-to-End Ownership & Requirements Translation: Independently scoped, designed, and deployed PKhata, converting an informal, non-technical business owner's requirements into a production system built around real local-market constraints and use cases.

Technical Ownership: Lead development of Aginiti-Redteam end-to-end, from architecture through implementation, as an open-source security research framework.

Cross-Team Collaboration: Partnered directly with clinicians at Kyung Hee University Hospital to translate real clinical requirements into KidneyCare's diagnostic logic.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Frameworks: Next.js, React, Node.js, Express

AI / ML: LLM Red-Teaming, RAG, LiteLLM, ChromaDB, Vector Search (PGVector), Embeddings, Multi-Agent Orchestration, Function Calling, Prompt Engineering

Backend & Cloud: PostgreSQL, Supabase, MongoDB, Docker, Vercel, Prisma ORM, Google OAuth

Security: OWASP Agentic AI Top 10, RBAC, Row-Level Security (RLS), Vulnerability Discovery